

### **7.01 Mathematical Preliminaries**

Learners are introduced to the fundamental categorisation of problems as existence, construction, enumeration and optimisation. They are also introduced to counting techniques that have a wide application across Discrete Mathematics.

Problems of conflict and cooperation are explored using game theory, including both pure and mixed strategies.

#### **Assumed knowledge**

Learners are assumed to know the content of GCSE (9–1) Mathematics and A Level Mathematics. They are also assumed to know the content of Pure Core